Sai Kiran Belana

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Education

University of Connecticut *Expected Graduation: December 2025*
Masters in Computer Science Engineering GPA: 3.7/4.0
Coursework: Systems Security, Operating Systems, Computer Architecture, Cyber Security, Analysis of Algorithms, Computer Networking, Big Data Analytics

Professional Experience

Summer Research Assistant, Goldenson Center, University of Connecticut *July 2025 - Present*
- Delivered a pricing model platform for insurance companies by architecting and integrating backend and frontend systems, improving financial projection accuracy and usability for stakeholders.
- Built a Python FastAPI backend to process mathematical models and a Next.js frontend with interactive profit/sales dashboards, reducing calculation time and enabling real-time scenario analysis.

Student IT Assistant, University of Connecticut *February 2024 - Present*
- Improved resource efficiency by 60% by containerizing university applications using Docker and deploying them on OpenShift Kubernetes clusters, enabling seamless autoscaling for dynamic workloads.
- Enhanced observability and performance monitoring by setting up and building Grafana dashboards to track key server metrics, allowing faster detection and resolution of infrastructure issues.
- Unified five classroom platforms into a single, scalable system with advanced filtering capabilities, reducing maintenance overhead by 80% and improving user navigation.
- Designed, developed, and deployed a OneDrive restoration tool using PHP and ReactJs; implemented CI/CD pipelines via Bitbucket and Jenkins to automate builds triggered by version control commits.

Software Engineer, LightSpeed Photonics *September 2021 - December 2023*
- Designed a RESTful platform with Slurm integration to deploy FPGA applications on Intel Arria 10-based HPC clusters remotely. Reduced manual overhead and streamlined hardware-software interaction for custom compute workflows.
- Accelerated compute performance by up to 10× by distributing matrix multiplication across FPGAs using OpenCL Channels. Benchmarked kernel performance against Xeon CPUs, enabling hardware-aware software optimization.
- Built a PCIe 3.0 profiling GUI to monitor data transfer between CPU and FPGA using FIFO buffers. Captured key metrics like bandwidth, LinkSpeed, and Link Width for round-trip performance analysis.
- Managed and provisioned cloud infrastructure across AWS, Azure, and GCP, including VPC networking, EC2 instance configuration, and S3 storage, ensuring high availability and scalability of deployed applications.

SWE Intern, LightSpeed Photonics *April 2021 - August 2021*
- Automated Yocto OS image builds using shell scripting, reducing built time by 75%.
- Built an E2E MERN application to demonstrate video analytics on custom EdgeAI Acceleration Hardware.

Projects

Secure Object Re-Identification and Trajectory Generation at the Edge *[Python, Trusted Execution Environments, WeaviateDB, Redis]*
- Building a real-time platform to securely perform object re-identification on traffic camera feeds by integrating Trusted Execution Environments at the edge, ensuring data privacy and compliance.
- Generated end-to-end object movement trajectories from origin to destination using WeaviateDB for vector storage and Redis for real-time metadata queuing, enabling faster analytics for traffic management.

WayNotify - A notification daemon for Wayland *[C++, CMake, Wayland, SDL2]*
- Developed a background service to listen for desktop notifications via D-Bus and display them on-screen using SDL2 and wlroots, achieving <10 ms latency per notification and full compatibility as a drop-in replacement for `dunst` on Wayland.

CPU Scheduling & File System Management Simulation *[Operating Systems, C++, Linux, Dear ImGui]*
- Built an interactive app using Dear ImGui library in C++ that invokes system calls to read/write files, monitor disk usage, modify files & set permissions and simulated job execution process of CPU scheduling algorithms(FCFS, SJF, SRTF, RR).

RISC-V Cache Architecture & Branch Predictor Implementation *[C, GCC, Assembly]*
- Constructed an out-of-order execution pipeline which supports 1-, 2-, and 4-way set associative caches.
- Implemented 1-level and 2-level dynamic branch predictors (2-bit) using BHT and BHSR tables.
- Compared performance metrics by testing set associativity and branch predictors to identify the best configuration.

More Projects

Skills

Programming Languages: C++ (proficient), Shell Script, PHP, JavaScript (prior experience)
Frameworks & Technologies: OpenCL, REST APIs, Electron, Node.js, React.js, SQL, NoSQL
Software Tools & Cloud: Git, Docker, Kubernetes, Jenkins, Slurm Workload Manager, Azure, AWS, GCP
Embedded & Hardware Integration: Linux, Arduino, ESP32, Yocto Project, RTOS, SPI, I2C, UART, Device drivers